

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – TECHNICAL PRESENTATION

Course Code:	CSA08	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 3 – Technical Presentation
Weightage:	10% of CIA	Total Marks:	100 (1 Question × 100 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	L Aravind	Reg. No.:	192424080
Section / Batch:	AI & DSA	Date:	

Instructions to Students

- Answer the given question through a technical presentation. Total Marks: 100.
- Clearly define the problem and explain the concept of algorithm growth functions.
- Present comparative analysis using appropriate representations such as graphs, tables, or charts.
- Use suitable tools (PowerPoint, visualization tools, or software) to support your explanation.
- Ensure technical accuracy, logical flow, and proper interpretation of results.
- Maintain clarity in communication, proper slide organization, and professional delivery.
- Time management, confidence, and audience engagement will be considered during evaluation.

Question Type	Focus Area	Questions	Marks
Technical Presentation	Analyze and compare growth functions using mathematical techniques and asymptotic concepts and with graphical and visual interpretation	Q1	100
GRAND TOTAL:		100 Marks	

SECTION A – Technical Presentation

Q1. Visualization-Based Technical Presentation on Growth Functions

Analyze growth functions $O(\log n)$, $O(n)$, $O(n \log n)$, $O(n^2)$, and $O(2^n)$ using visualization tools in a technical presentation format. Clearly define the purpose of comparing these functions for understanding algorithm efficiency. Use graphs, tables, or charts to show how each function behaves as the input size increases. Explain which functions remain practical for large-scale problems and justify your reasoning. Present the conclusions clearly and professionally by linking growth behavior to algorithm selection.

Analyzing Growth Functions

$O(\log n)$ · $O(n)$ · $O(n \log n)$ · $O(n^2)$ · $O(2^n)$

A visualization-based comparison of algorithmic growth rates and their impact on efficiency and scalability.

NAME L ARAVIND
 REGISTER NO. 192424080
 COURSE CSA06 · Design and Analysis of Algorithms
 CO ASSESSED CO1 — Precision (BL1, BL2, BL3)

SIMATS Engineering · Saveetha School of Engineering
 Department of Computer Science and Engineering



Why Compare Growth Functions?

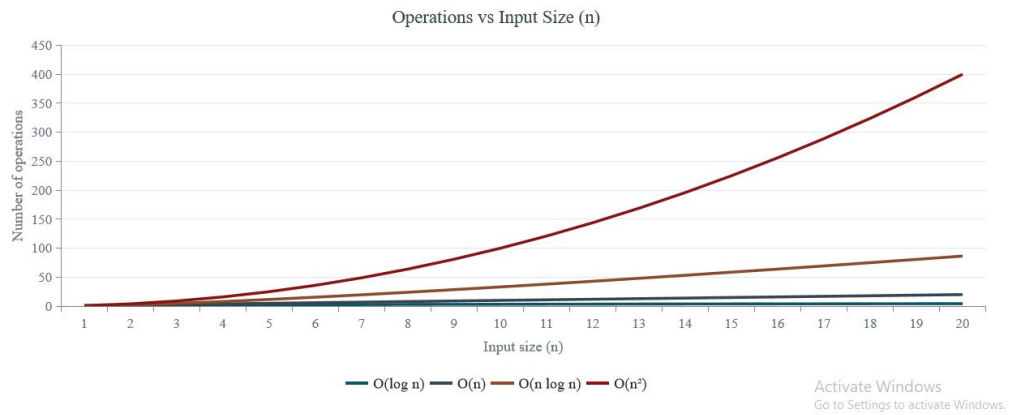


Asymptotic notation lets us predict scalability and compare algorithms fairly, independent of hardware — guiding which design remains usable as input size grows.

	Notation	Name	Example Algorithm	Behavior
	$O(\log n)$	Logarithmic	Binary search	Grows extremely slowly as n increases
	$O(n)$	Linear	Linear search	Grows directly proportional to n
	$O(n \log n)$	Linearithmic	Merge sort, Quick sort	Slightly steeper than linear, still efficient
	$O(n^2)$	Quadratic	Bubble sort, Insertion sort	Grows fast — doubling n roughly quadruples time
	$O(2^n)$	Exponential	Recursive subset generation	Grows explosively — becomes unusable quickly

Visualizing Growth: n from 1 to 20

$O(2^n)$ is excluded here — it dwarfs the other curves even by $n = 20$ (see next slide)



The Exponential Explosion

Why $O(2^n)$ cannot share an axis with the others

n	log n	n	n log n	n ²	2 ⁿ
5	2.32	5	12	25	32
10	3.32	10	33	100	1,024
15	3.91	15	59	225	32,768
20	4.32	20	86	400	1,048,576
25	4.64	25	116	625	33,554,432
30	4.91	30	147	900	1.07e+9

At $n = 30$, $O(2^n)$ already exceeds one billion operations — while $O(n^2)$ is under 1,000.

$O(\log n)$ EXCELLENT	$O(n)$ PRACTICAL	$O(n \log n)$ PRACTICAL	$O(n^2)$ RISKY	$O(2^n)$ IMPRACTICAL
--	---------------------------------------	--	-------------------------------------	---

Activate Windows
Go to Settings to activate Windows.

Which Functions Stay Practical?

Judging usability for large-scale, real-world input sizes

 $O(\log n)$	 $O(n)$	 $O(n \log n)$	x^1 $O(n^2)$	 $O(2^n)$
EXCELLENT Scales to billions of records with negligible cost increase.	PRACTICAL Cost grows steadily; fine for most large datasets.	PRACTICAL The best achievable for comparison-based sorting at scale.	RISKY Becomes slow beyond tens of thousands of items.	IMPRACTICAL Unusable beyond a few dozen inputs — avoid for large n .

Activate Windows
Go to Settings to activate Windows.

Conclusion

Linking growth behavior to algorithm selection

- ✓ Asymptotic notation lets us judge an algorithm's efficiency independent of hardware, letting the same analysis apply everywhere it runs.
- ✓ Logarithmic and linearithmic algorithms (binary search, merge sort) scale gracefully and should be preferred whenever the problem structure allows them.
- ✓ Quadratic algorithms are acceptable only for small or moderate n ; for large-scale systems they should be replaced with an $n \log n$ alternative.
- ✓ Exponential algorithms should be reserved for small inputs only, or replaced with dynamic programming, approximation, or heuristic techniques.
- ✓ Choosing the right algorithm is, fundamentally, choosing the right growth curve for the problem's expected input size.



Thank You — Questions & Discussion

Activate Windows
Go to Settings to activate Windows.

Code of Conduct

I, L. ARAVIND(192424080) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: L. Aravind

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Marks Evaluation:

Question No.	Technical Content Accuracy (30%)	Problem Identification (25%)	Use of Tools (20%)	Communication (15%)	Professionalism (10%)	Total Marks (100)
Q1	4	4	4	2	3	90
Overall Total (100)						90

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____